



Rhizo-PET: 4D PET system for analyzing carbon dynamics in the Rhizosphere

Wonu Park¹, Muhammad Nasir Ullah¹, Daniel Hastings², Craig Levin¹

¹Department of Radiology, School of Medicine, Stanford University

²Department of Environmental Science, University of California, Santa Cruz



Abstract

In this study, we investigate a novel approach to analyzing the Rhizosphere *in situ* with the use positron emission tomography (PET). Using ¹¹C, we can observe the carbon dynamics of the root structures of a *Phaseolus Vulgaris L.* plant and perform a 4D (x,y,z,time) analysis of such dynamics. Such 4D analysis involves drawing 1mm x 1mm x 0.5mm regions of interest (ROI) on the root structure along with surrounding soil structures and recording their time-activity curves (TACs), area under curve (AUC), and time-to-peak (TTP) graphs. We report the preliminary results of this method, which show a distinct kinetic profile of the carbon dynamics both on the overall root structure (from the top of the root to the tip of the root) and with fine structures adjacent to the root. A kinetic profile of a “hotspot” was also recorded, which show distinct differences in carbon influx/efflux compared to that of a root structure.

Introduction

The Rhizosphere- the region of soil surrounding the plant roots- mediates approximately 40% of the terrestrial carbon cycle through various processes. A better understanding of the root-soil interactions in the rhizosphere can provide crucial insights into improving agricultural practices, ecosystem management, enhancing soil health, improving plant productivity, and improving our knowledge of carbon sequestration and nutrient cycling.

Current methods for Rhizosphere analysis remain infeasible or inherently destructive. X-Ray and CT remain viable due to the soil’s density and homogeneity. Extracting the roots from the soil for analysis interferes with or destroys fine structures needed for precise analysis

We developed a novel method of imaging using positron emission tomography to help overcome these difficulties in live analysis of the rhizosphere, using ¹¹CO₂ as our radionuclide emitting tracer.

Figure: Our data acquisition, processing, and reconstruction pipeline.

Materials and Methods

System

- 8 Detector modules controlled by 2 EDAC-16 units
- Each detector module is 50mm x 50mm Hamamatsu 8500 PSPMT, 48 x 48 mm² active LYSO scintillator array

Method

- ¹¹CO₂ was introduced to the *Phaseolus Vulgaris L.* leaves for 20 minutes before being removed, allowing the plant to photosynthesize.
- The plant was placed on a rotation platform, allowing the detectors to pick up what would otherwise be dead zones.
- All data was collected on coincidence mode.
- Data was processed, and faulty data was removed, stored in Listmode.
 - No scatter correction or attenuation was applied
- Listmode Data was separated into 5 minute bins.
- CUDA reconstruction using voxel sizes of 0.5mm, 0.5mm, 0.5mm, and 40 iterations for each bin.
 - Normalization files were used, gathered from previous data collection sessions.
- A variety of python scripts were used for post processing and visualization purposes

Experimental Setup

For three parts of the plant (Top, middle, and bottom), 9 ROIs were drawn. the center ROI is the root structure, while other ROIs represent the activity levels some distance away from the root. For all ROIs, TACs were created by calculating the mean pixel value of the 2px x 2px x 1px ROI for each time frame, and plotting the mean activity over time. A gaussian smoothing was applied for all TACs.

Results

The time-activity-curves and the time-to-peak curves for a root structure is drawn to the side. The top left graph shows the TACs and its first derivative for the top of the root, the top right graph shows the middle root, and the bottom left graph shows the bottom root. The Time-to-peak graph at the bottom right shows a distinct pattern of carbon flow moving down the plant with TTP=50 minutes for the top root and TTP=60 minutes for the bottom root. Also seen is the fact that the soil ROIs adjacent to the root tend to peak later than their respective roots, indicating further carbon dynamics from the root structure to fine root hairs.

Conclusion

Despite missing scatter and attenuation correction, Rhizo-PET has already shown to be a robust method for Rhizosphere imaging. Kinetic profiles can help distinguish between root structures and potential “hotspots.” Hotspots can be further analyzed by uprooting the plants and investigating the region of interest physically.

These results help advance our understanding of the *in situ* plant rhizosphere. This has profound impacts for understanding soil carbon sequestration, global food security, climate resilience, as well as improving soil health and plant productivity.

Further Works

Further works can be seen through the implementation of scatter correction and attenuation. These will be needed to make precise statements about the processed and visualized data. Additionally, artifacts seen in CUDA reconstruction can be mitigated via alternate methods of reconstruction such as parallelproj reconstruction (image seen below). Finally, different temporal bin sizes could potentially provide more accurate data and be useful for more precise analysis of carbon dynamics.

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